

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Farmers often struggle to choose the most suitable crop due to varying soil and environmental conditions. Manual analysis of factors such as Nitrogen (N), Phosphorous (P), Potassium (K), pH, temperature, humidity, and rainfall can be difficult and inaccurate. OptiCrop: Smart Agricultural Production Optimization Engine uses machine learning to recommend the best crop, improve productivity, and support sustainable agricultural practices.

Problem Statement (PS)	I am (Customer)	I'm trying to	But Because	Which makes me feel
PS-1	A farmer	Select the best crop for my field to maximize yield and profit.	I don't have enough knowledge about soil nutrients, weather conditions, and crop suitability.	Uncertain about making the right farming decisions.
PS-2	A farmer	Check whether my current soil and climate conditions are suitable for a particular crop.	It is difficult to analyze multiple environmental factors like N, P, K, pH, temperature, humidity, and rainfall manually.	Confused about the productivity and success of the crop.
PS-3	An agricultural researcher or policymaker	Analyze crop-environment relationships to improve agricultural planning and sustainability.	Agricultural data is vast and complex, making it difficult to identify meaningful patterns without intelligent tools.	Challenged in making accurate, data-driven decisions for sustainable farming.